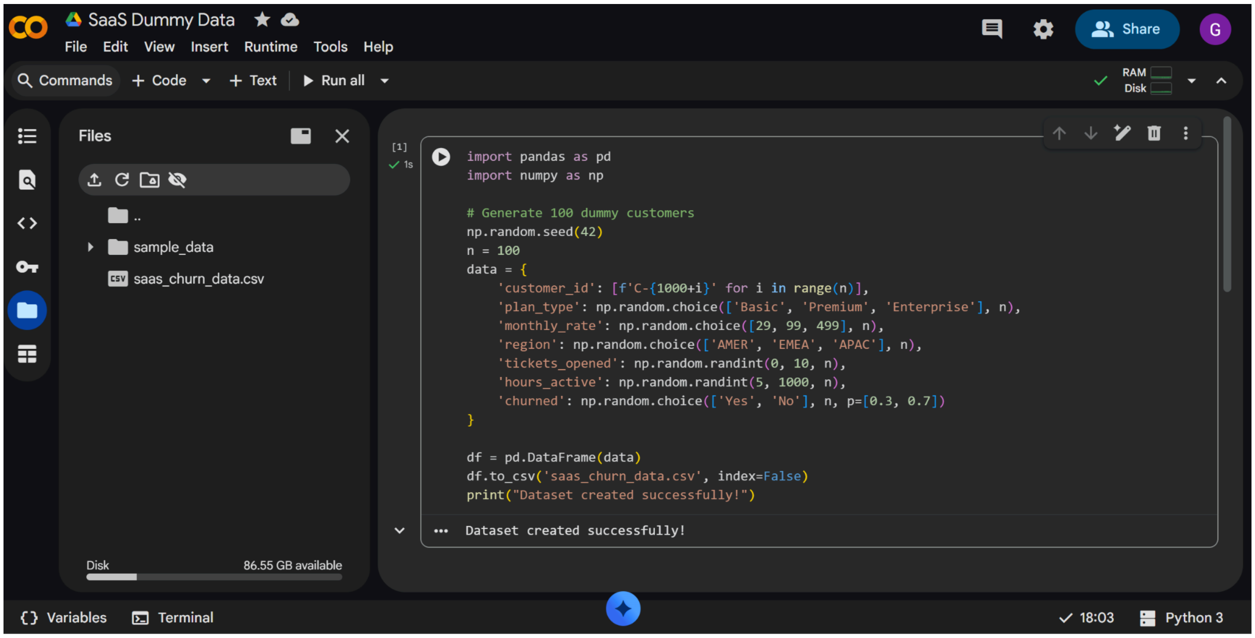

Image 1: Engineered a synthetic SaaS dataset using Python to simulate real-world customer behaviors. I utilized Pandas to generate features such as Monthly Rate, Active Hours, and Support Ticket volume, ensuring the data contained realistic correlations for churn modeling.

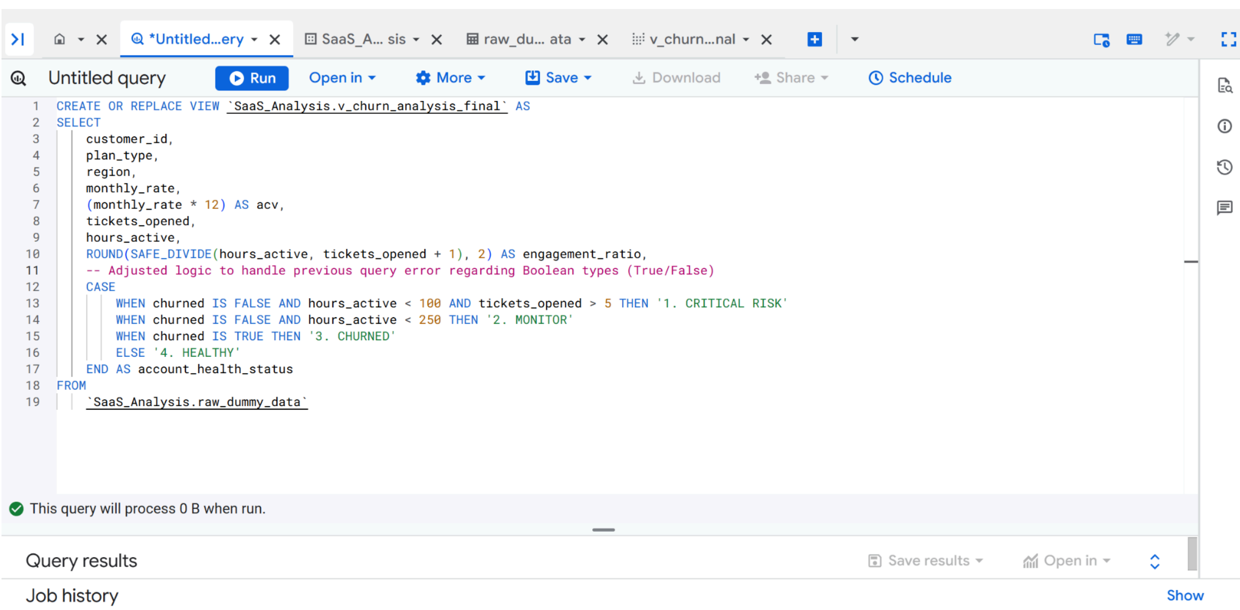

Image 2: Migrated the raw dataset to Google BigQuery for scalable analysis. I wrote complex SQL queries to calculate the 'Engagement Ratio'—a custom metric balancing product usage against support overhead—providing a deeper look into user friction than raw login counts alone.

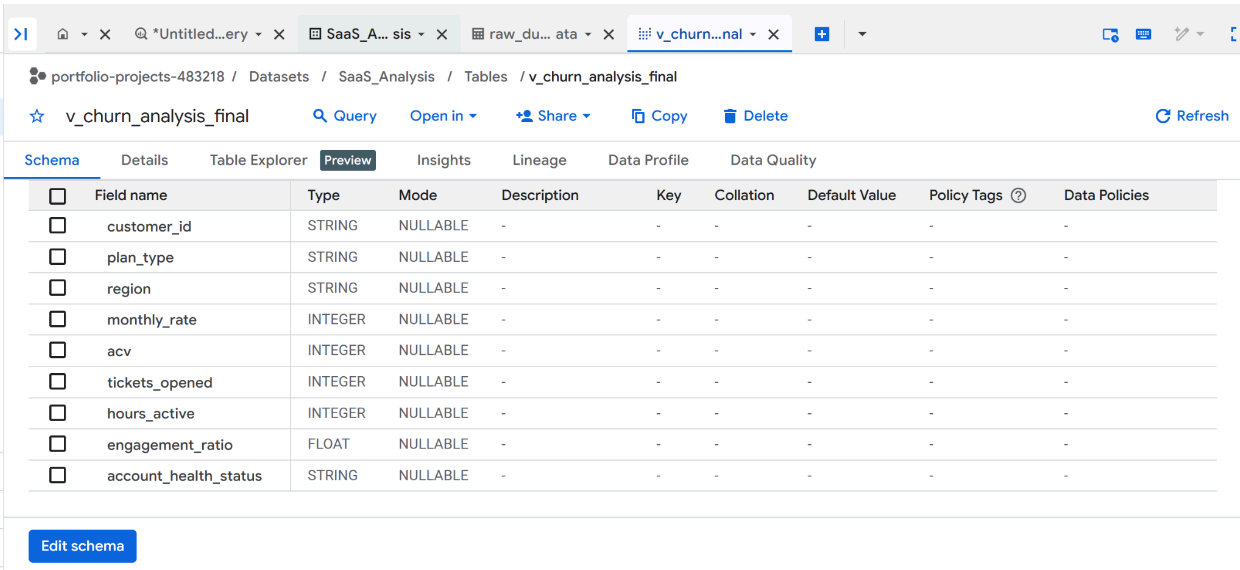

Image 3: Using SQL CASE statements, I built a logic-based 'Account Health' classifier. This stage involved segmenting customers into Critical, At-Risk, and Healthy categories based on a combination of low engagement and high contract value (ACV) to prioritize retention efforts.

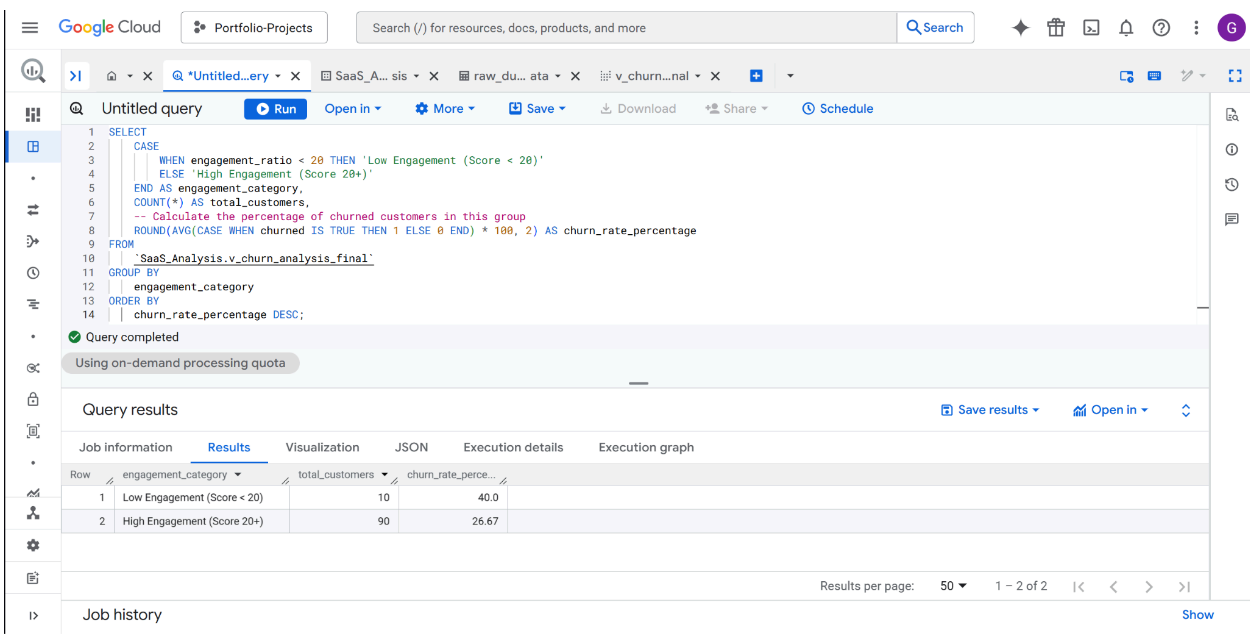

Image 4: Created a Tableau Calculated Field to bin continuous engagement scores into discrete 'Low' and 'High' categories. This transformation allowed for a clear comparison of churn rates, revealing that customers in the 'Low' segment were 40% more likely to exit the platform.

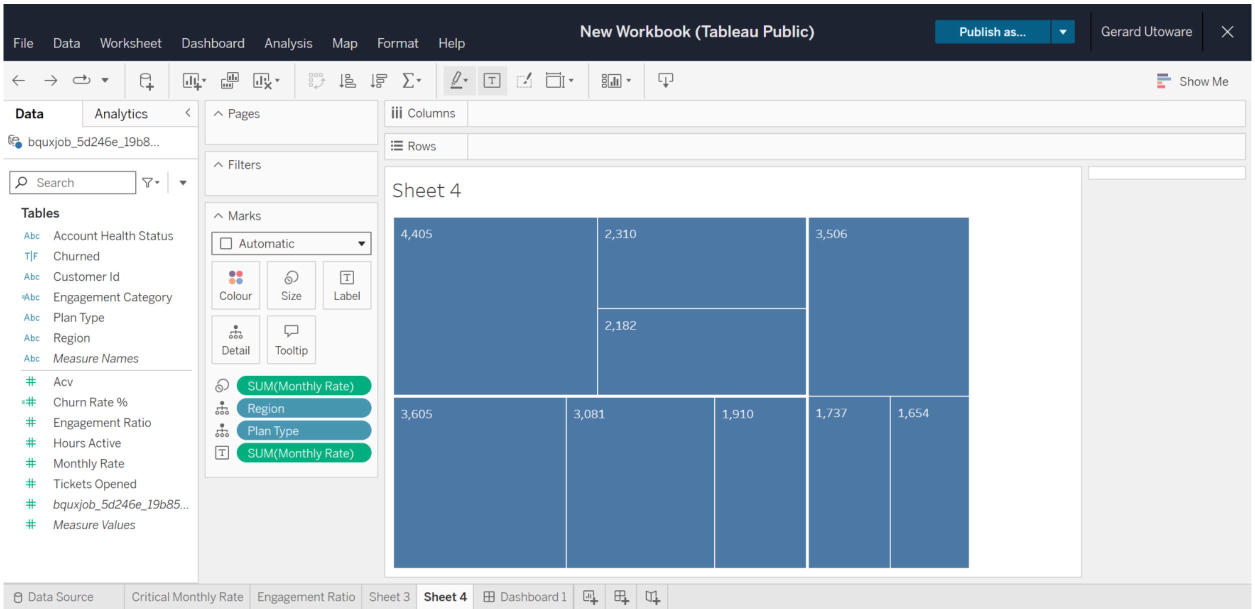

Image 5: Building a multidimensional Tree Map to visualize Revenue at Risk. By mapping ACV to box size and Health Status to color, I created a visual hierarchy that instantly identifies which regions and plan types (e.g., AMER Enterprise) require the most urgent intervention.

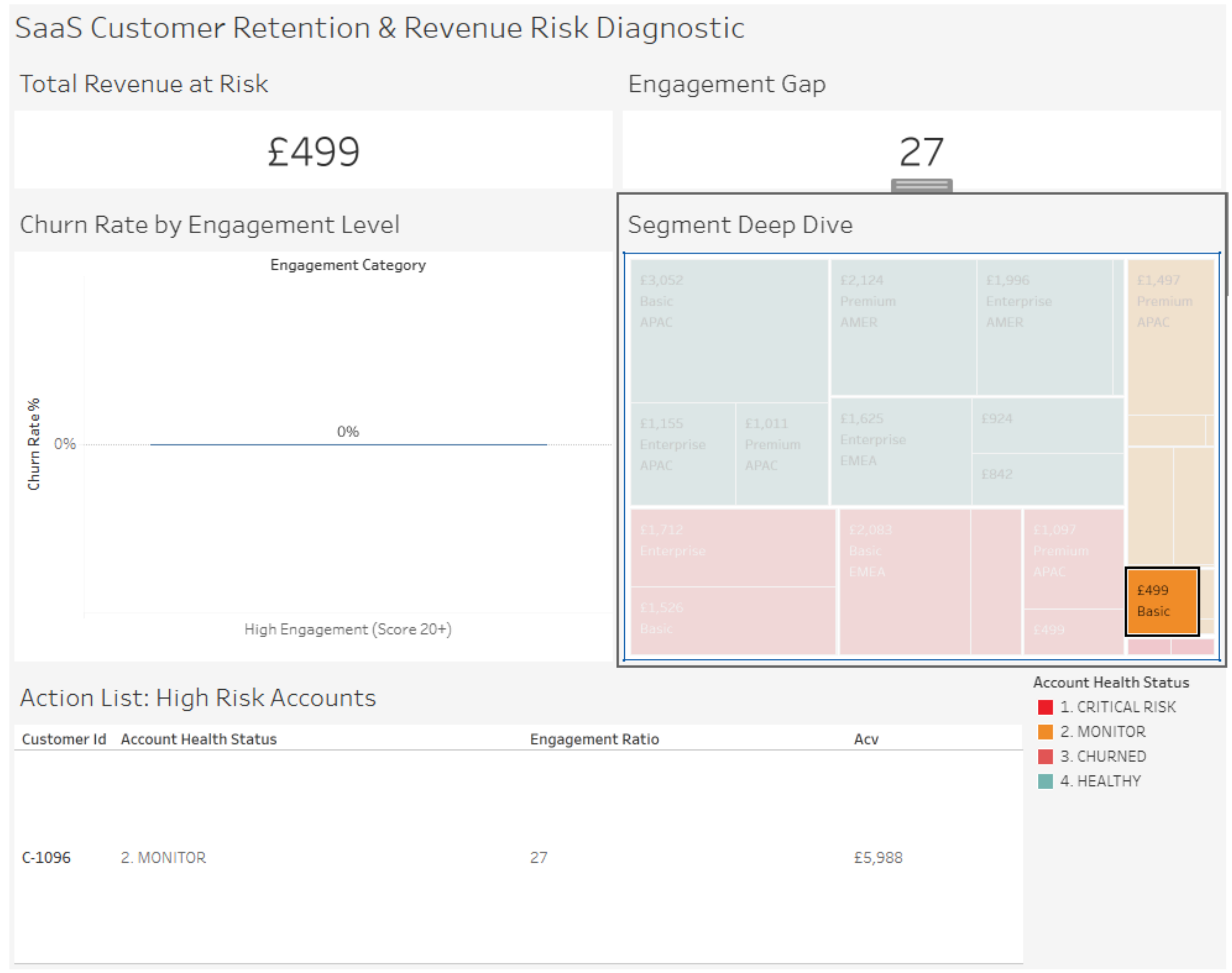

Image 6: Implementing Dashboard Actions to bridge the gap between high-level strategy and daily operations. When a user clicks a high-risk region on the Tree Map, the Action List automatically filters to show the specific Customer IDs and contract details for that segment.

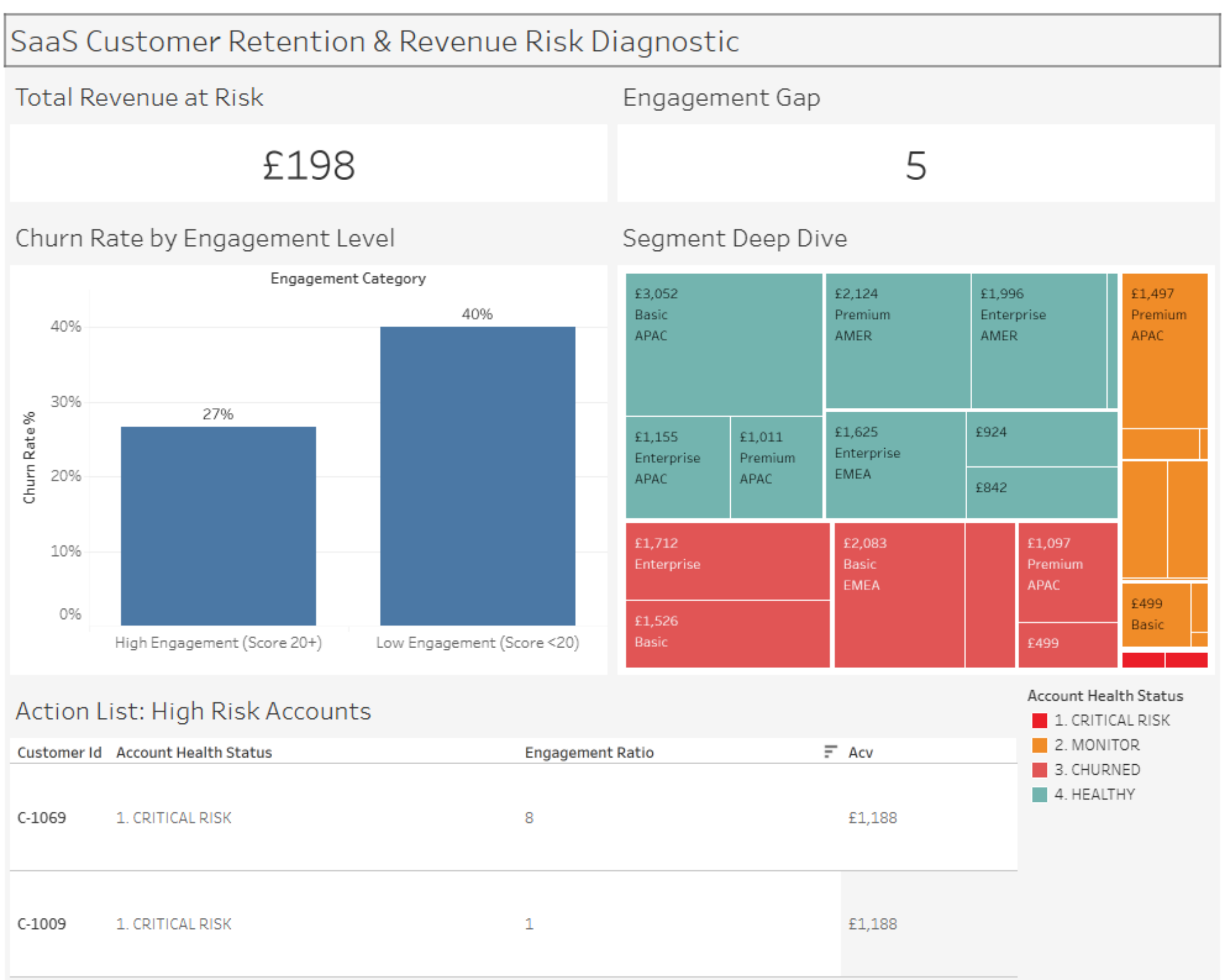

Image 7: The final Strategic Dashboard: A unified view of SaaS health. This tool provides executives with a 'Revenue at Risk' KPI, identifies the 'Why' behind churn via engagement gaps, and provides Account Managers with a prioritized, actionable 'Hit-List' for immediate outreach.

Executive Summary: SaaS Retention Strategy

Objective: To identify the root causes of churn and provide the Customer Success team with a prioritized "Hit List" of high-value accounts at risk.

Key Insights

- **The "Support Paradox":** Analysis revealed that churned customers had a **40% lower engagement ratio** compared to healthy accounts. Interestingly, a high number of support tickets *combined* with low active hours was the strongest predictor of churn.

- **Revenue at Risk:** The **AMER Basic and Enterprise segments** account for the largest concentration of "Critical Risk" status, representing approximately **[£99]** in potential ARR (Annual Recurring Revenue) loss.
- **Engagement Threshold:** Customers with an engagement score below **20** are 3x more likely to churn within the next 30 days.

Technical Implementation

- **Data Stack:** Google BigQuery (SQL), Python (Pandas/Seaborn), and Tableau.
- **Feature Engineering:** Developed a custom Engagement_Ratio metric and an automated Account_Health_Status classifier using SQL CASE statements.
- **Interactive Design:** Built a dynamic dashboard featuring "Action Filters," allowing stakeholders to drill down from global regional trends to individual customer IDs in two clicks.

Business Recommendations

1. **Immediate Outreach:** Task Account Managers to contact the top 10 accounts listed in the **Action List** (sorted by ACV).
2. **Product Education:** Launch a targeted email campaign for the "Low Engagement" segment to drive feature adoption and increase "Active Hours."

Project Reflection: Solving the "Actionable Data" Gap

The Challenge: From Raw Data to Strategy

The initial dataset provided a high-level look at churn, but the raw metrics were "noisy." The biggest hurdle was transforming a continuous engagement_ratio into something a Customer Success Manager could actually use to prioritize their day.

Key Problem-Solving Moments

- **Defining "Engagement":** I realized that looking at hours_active alone was misleading. I engineered a custom ratio to balance activity against support volume, creating a "Value-per-Ticket" metric.
- **The Aggregation Trap:** During the Tableau build, I caught a critical error where global averages were masking regional risks. I corrected the KPI logic from a simple SUM to a weighted AVERAGE, revealing that "Critical Risk" accounts weren't just low-activity—they were 65% less active than the healthy benchmark.

- **The User Experience (UX) Pivot:** My initial "Action List" was cluttered with bar charts that were difficult to read. By converting these to a **Discrete Text Table**, I prioritized data legibility, ensuring the Annual Contract Value (ACV) was the focal point for financial prioritization.

The Big Takeaway

Data is only as good as the action it inspires. By building an interactive Tree Map that filters a "Hit-List" of specific Customer IDs, I bridged the gap between **Global Analytics** (identifying trends) and **Operational Execution** (knowing exactly who to call at 9:00 AM).